Russian Sign Language Dactyl Recognition

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***Abstract*—In this paper, we compare several real-time sign language dactyl recognition systems and present a new model based on deep convolutional neural networks. These systems are able to recognize Russian alphabet letters presented as static signs in Russian Sign language used by people from deaf community. In such an approach, we recognize words from Russian natural language presented by consequent hand gestures of each letter. We evaluate our approach on Russian (RSL) sign language, for which we collect our own dataset and evaluate dactyl recognition.**

***Keywords*—Sign Language Translation; Russian Sign Language; Hand Gesture Recognition; Deep Convolutional Neural Networks.**

I. Introduction

Sign language is a communication tool for people with hearing or speech impairments, denoted by deaf community. In Russia, its size is estimated as 9 million people. Among different disabilities deafness is one of the most frequent, however, it can mostly be observed by how deaf people communicate. Most of them do not consider inability to hear or speak as a disability rather as a boundary in communication and specific experience, which they share with their friends using the language of signs, gestures and emotions.

Nowadays, computer vision and natural language processing via deep neural networks succeeded in many applications, including machine translation and gesture recognition. Thus, many researchers try to combine recent models to make people from deaf community communicate more easily using AI-powered solutions in real-world scenario.

Sign language recognition is usually step-wise process combining several machine learning models in common pipeline. Starting with dataset collection (Web, phone camera, depth sensor) and its labelling, machine learning models are build to extract features by training model for alphabet letters classification based on image of hand gesture. For the trained model, we measure its performance according to classification quality metrics evaluating on test dataset.

In what follows, we overview existing solutions, describe several models and evaluate them in Russian sign language dactyl recognition.

A. Recognition using Sensors

Using additional sensors to RGB image, such as depth sensor, gyroscope, accelerometer may improve overall spatial information and leads to better performance of recognition models as trade-off of time complexity of algorithms performing fusion of RGB and other information in combined model.

Hand gestures may be done via the phone moving, so accelerometer data with start and end time points of gestures may be used for gesture recognition [1] and gesture curve estimation [2]. The limitations are necessity of network connection, gadget and not accurate prediction of sensor position.

B. Recognition RGB Images

One of the early attempts to study hand gesture recognition were based on colour histogram distributions or skin detection. Hand skin detection algorithms [3], Viola-Jones [4] or HOG [5] filtering, dimensionality reduction approaches [6] showed good performance but were dependent on manual feature engineering. Standard methods for hand gesture classification using SURF [7], SVM kernel-based models [8], kNN classification [9] also suffer from not accurate prediction.

Now, convolutional neural networks are trained to learn proper filters without manual feature engineering under loss function optimization chosen for recognition task, while also having the ability to apply trained model for data different from train domain [10].

Moreover, fusion of RGB and depth sensor information is far more easy to do with deep convolutional neural networks [11], [12]. Adding temporal component to such models leads to proper continuous sign language recognition [13], [14].

In our work, we consider convolutional neural networks with simple structure of several convolution and pooling layers with ReLU activations and dense layers in certain architectures. We train them for static sign recognition under memory and time constraints comparing to the results of [15], [16].

II. Sign Language Dactyl Description

Sign Language (SL) consists of hand gestures together with facial expressions and body position, being either dynamic or static by nature of sign gesturing. Important part of SL gesture for name entities is dactyl representing words by consequent letters, each translated by corresponding signs.

Russian Sign Language (RSL) is a language of deaf community in Russia and some of its neighboring countries. Russian language is an official language of United Nations. Presenting people leaving on large territories world-wide RSL attracts interest of researchers. RSL dactyl contains 33 gestures, only 26 gestures are static or can be named as “static”, they are presented in Table I. Seven other gestures are dynamic and not observed in our main dataset [17] for RSL.



Figure 1. Dactyl Samples from RSL

III. Datasets and Model

Since [18], we improved our dataset [17] and collect more real-world samples for the recognition system for Russian alphabet collected as the static images (gestures that can be observed as static, for example, "c" ("ц") and "k" ("к") are also named as "static" gestures).

The list of observed RSL gestures represents gestures for the letters of Russian alphabet from Table I.

The data for RSL dactyl were gathered via processing youtube videos with sign language video lessons and by asking people submit their videos of signing letters via created web portal [1]. As a consequence, the dataset contains real-world images and videos processed by trained CNN for hand detection. It is important for the model to take into account only valuable features, while learning to ignore other image conditions, such as person clothes and skin colour, lightning or shadows.

Our dataset contains both, high resolution iamges taken using smartphones and low resolution images obtained from decomposing youtube videos. RSL image samples are shown in Figure 1. The dataset is located on the GitHub repository [17]. The repository README.txt lists the source links for the YouTube videos and images' name decoding algorithm. We intent to make the dataset open for non-commercial use and make it a part of our contribution.

A. Sign Language Dactyl Recognition using CNN

In this paper, we evaluate enlarged (compared to [18]) RSL dataset and compare two models suggested in [18] for dactyl recognition with models presented in [15], [16]. Our models take 32×32 images as an input, while other two models take as input grayscale images of size 28×28. We chose a larger size improving prediction quality while maintaining comparable performance and parameter space. The detailed architecture of the all considered model is shown in Figure 2.

We take LeNet-like [19] for the first architecture applied on RGB images with two convolutional layers with 5×5 kernel and ReLU as an activation function, followed by MaxPooling layer with filters of size 2×2 applied with a stride of 2. The resulting features are then being input of 2 dense layers with ReLU activation and a SoftMax predictions for each letter class. We aimed on fast recognition algorithm, so reducing feature spaces via MaxPool layer and small filter sizes outputs helps us achieve our goal while preserving quality of predictions.

Next, we take "QuadroConvPoolNet" [18] architecture providing similar model to [20], but with different pooling positions and different filters. The input of the model is grayscale image which helps us get high processing speed while having low memory consumption suitable for mobile devices.

We compare our models with the models presented recently for real-time ASL dactyl recognition [15], [16]. The second model was pretrained encoder part, which is not well applied for real-world noisy images in RSL dataset.

We trained all networks on 100 epochs using Adam optimizer [21] with a learning rate of 0.003, which appear to outperform other optimizer algorithms on our datasets in terms of training speed. Cross-entropy function was used for our multi-class classification task.

IV. Experiment Results

Sign Language dactyl recognition is a variant of multi-class classification, in which eash letter as standalone class. The input images are taken from RSL static gesture image dataset labelled according to correct sign gesture. For more robust training, we augment our dataset by rotating them on random turns of $\{-16, -10, 0, 10, 16\}$ degrees. Thus from around 900 original images 3500 new images were obtained.

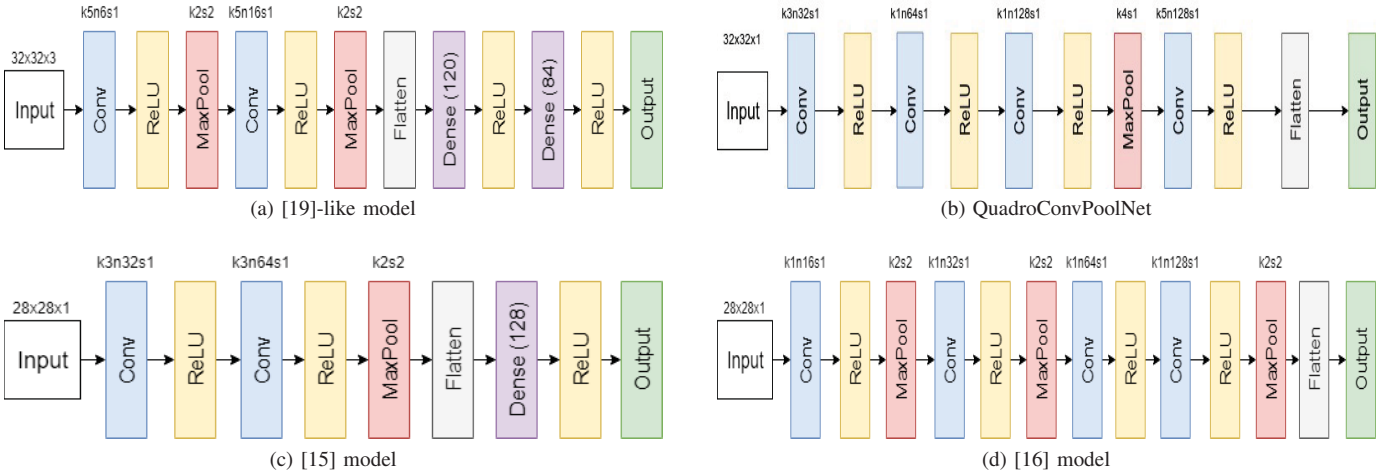
We measure accuracy for multi-class classification to measure quality representing a number of correct predictions over all predictions made. We validate classification via N binary classification of one letter versus all others,where N is number of classes or just number of static signs. Our classifier approximates probability for each class via SoftMax $\sigma(z)_i = \frac{e^{z_i}}{\sum_{n=1}^{N} e^{z_n}}$. Based on this values, the class with the maximal estimated values is chosen as prediction result, in which the model is the most confident. In order to measure the model quality one need to compute average number of proper predictions $\frac{1}{Z} \sum_{z=1}^{Z} [y_{pred} = y_{true}]$. Loss function for multi-class classification was used as categorical cross-entropy measuring quality by multi-class binary classification.

We train our model using random shuffle of augmented images of each class and split them into Train and Test sets with the ratio 80%/20%. Interpreting results on Test dataset we see that even compact architectures of CNNs may work better than the complex one trained on laboratory data, but having problems to generalize on real-world dataset. The results on RSL varies based on letter classes having with the worst indicators: 70-80% accuracy, however, we belive that enlarging our dataset will make it possible to have accuracy over 98%

[1] https://sl-data.ddns.net/

Table I. RUSSIAN ALPHABET AND ITS TRANSCRIPTIONS. “x” (“y”) MEANS THAT y IS A RUSSIAN LETTER AND x BEING ITS ENGLISH TRANSCRIPTION

“a” (“а”)	“b” (“б”)	“c” (“ц”)	“ch” (“ч”)	“e” (“е”)	“f” (“ф”)
“g” (“г”)	“i” (“и”)	“k” (“к”)	“kh” (“х”)	“l” (“л”)	“m” (“м”)
“n” (“н”)	“o” (“о”)	“p” (“п”)	“r” (“р”)	“s” (“ф”)	“sh” (“ш”)
“t” (“т”)	“u” (“у”)	“v” (“в”)	“ya” (“я”)	“ye” (“э”)	“yi” (“ы”)
“yu” (“ю”)	“zh” (“ж”)				



(a) [19]-like model (b) QuadroConvPoolNet

(c) [15] model (d) [16] model

Figure 2. **Architecture of models**. Each convolutional layer (as well as MaxPool layer) has detailed description of the structure: the kernel (filter) size (k), number of filters (n), and strides (s).

for all the letters. Details of models accuracy for each letter are shown in Table II.

Table II. ACCURACY ON RSL TEST DATA

	[LeCun, 1998] model	QuadroConv-PoolNet	[Taskiran et al. 2018] model	[Chakraborty et al. 2018] model
a	51%	70%	60%	20%
b	65%	75%	68%	22%
c	25%	87%	50%	25%
ch	82%	85%	85%	43%
e	75%	69%	68%	40%
f	85%	82%	85%	30%
g	68%	84%	59%	39%
i	83%	69%	78%	40%
k	83%	86%	84%	33%
kh	55%	80%	60%	45%
l	65%	78%	81%	29%
m	78%	77%	80%	27%
n	76%	82%	60%	24%
o	79%	80%	81%	21%
p	65%	61%	60%	20%
r	67%	88%	72%	45%
s	89%	93%	87%	35%
sh	62%	62%	60%	55%
t	68%	69%	65%	65%
u	82%	88%	83%	40%
v	75%	86%	70%	27%
ya	76%	81%	83%	32%
ye	58%	67%	50%	20%
yi	73%	65%	70%	26%
yu	69%	69%	68%	30%
zh	70%	83%	71%	28%
Overall	73%	78%	71%	33%

V. DISCUSSION

We enlarged our own RSL dataset [17] and improved results on RSL dactyl recognition. We showed that even simple architectures can outperform more complex ones ([16] when trained on real-world images, which are far harder to train on compared to ASL laboratory dataset. We aim to reduce the influence of various image shooting factors, such as lightning, clothes, shadows and background conditions, while switching our model from image recognition to video recognition for datyl data, which we collect as a part of RSL dataset. We also aim to compare our approach with greater number of hand gesture recognition methods presented in code implementation aggregators on GitHub[2],[3].

VI. CONCLUSION

We compared two compact CNN models to recognize RSL dactyl static hand gesture images. We collected our own RSL dactyl dataset and train the model on raw images and obtained much better quality compared to [18].

In contrast to many studies, we aimed on comparison on recognition systems trained in “ideal” conditions and real-world datasets that we collected. This resulted are the first sufficient results of RSL dactyl recognition.

[2]https://github.com/xinghaochen/awesome-hand-pose-estimation
[3]https://github.com/ChanChiChoi/awesome-Gesture-Recognition

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The results reported in this paper improve the results from ACM PETRA 2019 conference [18], in which we achieve state-of-the-art performance on American SL dataset, but fail to capture RSL dactyl due to lack of data. Now, we received results comparable to other laboratory datasets on SL dactyl recognition for Russian language as well.

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